

Week 2 Progress Report: Literature Review on SIEM Systems and Log-Based Detection Techniques

Introduction

Security Information and Event Management (SIEM) systems are foundational technologies in modern cybersecurity operations used to collect, aggregate, normalize, and analyze security-relevant logs and events from a wide range of sources. SIEM platforms enable organizations to detect, investigate, and respond to malicious behavior by providing centralized visibility and real-time threat detection capabilities. These systems play a critical role in helping security teams identify threats that would otherwise remain hidden in isolated or uncorrelated log data (González-Granadillo et al., 2021).

Overview of SIEM Systems

SIEM systems are designed to provide a unified view of an organization's security posture by ingesting log and event data from disparate sources such as operating systems, firewalls, intrusion detection systems, and applications. These platforms perform log aggregation, parsing, normalization, correlation, and alerting to identify suspicious activity across an environment (ScienceDirect, n.d.).

González-Granadillo et al. (2021) describe a typical SIEM architecture as consisting of several core components, including log collection mechanisms, normalization engines, correlation logic, data storage, and monitoring interfaces. Each of these components is essential for enabling effective detection and analysis. Without proper normalization and correlation, raw log data provides limited security value.

Industry sources further emphasize that SIEM systems extend traditional log management by enabling real-time monitoring, security analytics, and incident response workflows. This combination of long-term log storage and real-time analysis distinguishes SIEM platforms from basic logging tools (Palo Alto Networks, n.d.).

Log-Based Detection Techniques

A core capability of SIEM systems is log-based detection, which relies on analyzing event data to identify known or suspicious behaviors. One of the most widely used techniques is rule-based detection, where predefined rules identify patterns such as repeated failed login attempts, unauthorized access attempts, or privilege escalation events (Exabeam, n.d.).

In addition to simple rule matching, SIEM platforms commonly employ threshold-based and time-window analysis. For example, a large number of failed authentication attempts within a short time period may indicate a brute-force attack. Correlation techniques further enhance detection by linking related events across different log sources, such as authentication logs combined with system or audit logs. This correlation improves detection accuracy and reduces false positives by providing greater context (Exabeam, n.d.).

While some modern SIEM systems incorporate machine learning and behavioral analytics, existing literature notes that rule-based detection remains foundational due to its explainability, reliability, and effectiveness in identifying known attack techniques.

Role and Importance of Audit Logs

Audit logs are a specialized category of logs that record detailed information about user actions and system-level events, including command execution, file access, privilege changes, and configuration modifications. An audit trail provides a chronological record of activity that allows security analysts to reconstruct events and investigate suspicious behavior (Wikipedia, n.d.).

Research highlights the importance of audit logs for detecting insider threats, privilege escalation, and persistence mechanisms that may not be visible through standard authentication or system logs alone. Properly managed audit logs are also critical for forensic investigations and compliance requirements, as they provide reliable evidence of system activity (Ali et al., 2021). Including audit logs in SIEM analysis significantly enhances visibility and accountability by allowing analysts to trace actions back to specific users or processes. This makes audit logs a valuable data source for both detection and post-incident analysis.

Challenges in SIEM Implementation

Despite their effectiveness, SIEM systems face several challenges identified in both academic and industry literature. These challenges include high log volumes, difficulty in normalizing heterogeneous log formats, and the potential for alert fatigue caused by poorly tuned detection rules (Ali et al., 2021). Additionally, the effectiveness of a SIEM is highly dependent on log quality and completeness; missing or inconsistent logs can lead to blind spots in detection.

Industry frameworks emphasize the importance of clearly defining logging requirements and detection goals before SIEM implementation. Without a purpose-driven approach, SIEM deployments may become overly complex and difficult to manage (ISACA, 2023).

Relevance to Capstone Project Design

The findings from this literature review directly inform the design of the proposed lightweight SIEM project. Specifically, the review highlights the importance of centralized log collection, normalization into a unified event schema, rule-based and threshold-based detection, and the integration of audit logs to improve visibility and accountability. Additionally, understanding the limitations of SIEM systems helps establish realistic expectations for detection effectiveness and provides a foundation for evaluating system performance later in the project.

Key Takeaways and Preparation for Week 3

This literature review provided a foundational understanding of how Security Information and Event Management (SIEM) systems operate and the role they play in modern security operations. Reviewing both academic and industry sources clarified that effective SIEM platforms rely on a small number of core components, including centralized log collection, normalization of heterogeneous data, rule-based detection, event correlation, and alerting. Understanding these components helped reinforce that a SIEM's effectiveness is determined not by the volume of data collected, but by the quality of log sources and the logic used to analyze them.

One key takeaway from this review is the importance of explainable, rule-based detection techniques. While advanced methods such as machine learning are increasingly discussed in the literature, rule-based and threshold-driven approaches remain the foundation of most SIEM deployments due to their transparency and reliability. This insight supports the decision to focus on detection engineering rather than attack execution in this project.

Another important takeaway is the value of audit logs as a complementary data source to traditional security logs. Audit logs provide deeper visibility into user actions, privilege changes, and system modifications, which are critical for detecting behaviors such as privilege escalation and persistence. Integrating audit logs into the SIEM design will improve detection accuracy and support accountability and forensic analysis.

These findings directly inform the goals for Week 3 of the project. The next phase will focus on translating these concepts into a concrete SIEM architecture by defining system components, designing a unified event schema, and outlining how security and audit logs will be ingested, stored, and analyzed. The literature review ensures that design decisions made in Week 3 are grounded in established research and industry best practices rather than ad hoc implementation choices.

References

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